Research Project: Franciscan Eco-Theology and Environmental Attitudes

A term paper submitted for the
RM in Management and Applied Microeconomics

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Abstract

Our project focuses on the effect of exposure to the main Mendicant orders of the 12th and 15th century - the Franciscan and Dominican - on the contemporary environmental attitudes. We find that that higher Franciscan is associated with less pro-environmental attitudes, while higher Dominican exposure is associated with more pro-environmental attitudes.

Contents

1 Introduction

Religious institutions, in particular the Church, played a pervasive role in every aspect of life in medieval Europe, shaping people's lives, values, and behavior throughout the Middle Ages (Lynch, 2014). One of the most famous religious figures of the Medieval Ages is St.Francis of Assisi, who is the founder of Franciscan order. Being recognized by the Catholic Church as the patron saint of ecology, he is widely known for his deep affection towards nature and animals.

In this research, we study Franciscan eco-theology and its effect on contemporary environmental attitudes. We have also controlled for the Dominican order, which is the other famous Mendicant order of the Medieval Ages with comparably equal historical importance. Thus, our research question is the following: What is the effect of exposure to Franciscan "Eco-Theology" from the 1200s to the 1500s on contemporary environmental attitudes among European populations? (compared to Dominican theology).

To do so, we first construct our independent variable - the historical exposure measures for both Franciscan and Dominican orders, defined as a fraction of a region's historical population living geographically close to a monastic house. We then construct our dependent variable - environmental attitudes index - by applying Principal Component Analysis to the individual-level survey data on environmental attitudes from the European Values Study (EVS) 2017–2021 dataset. Using those variables and the set of controls, we empirically examine how historical exposure to the two Mendicant orders affects contemporary environmental attitudes.

Given the historical background of St. Francis of Assisi, we have set the following initial hypothesis: we expected that higher historical exposure to the Franciscan Order is associated with more pro-environmental attitudes, while higher historical exposure to the Dominican Order is associated with less pro-environmental attitudes. However, our empirical results show the opposite. Contrary to our initial expectations, we find that higher historical exposure to the Franciscan Order is associated with less pro-environmental attitudes,while higher historical exposure to the Dominican Order is associated with more pro-environmental attitudes.

2 Context and Hypotheses

The Franciscan Order was founded in the early 13th century by St. Francis of Assisi (1181-1226), who was recognized by the Catholic Church as the patron saint of ecology, widely known for his profound affection towards nature and animals. This deep affection towards animals and nature can be seen in his "Canticles of the Creatures," where St. Francis refers to the sun and the moon as brother and sister. Importantly, Francis and his followers emphasized the close imitation of life of Christ and apostles through the ideals of poverty and humility. "He wanted to embrace real poverty, without any rationalisations or modifications. His band of followers would beg for their daily food, would take no care for tomorrow, and would own nothing – really nothing." (Lynch, 2014, p. 258). Thus, the Franciscan Order focuses on collective life, communal care, antimateralism, shame, compassion, simplicity, and humbleness (Arrunada and Lopez-Manuel, 2024).

In contrast, the Dominican Order, founded by St. Dominic (1170-1221) in the early 13th century, prioritized intellectual and educational mission of medieval Catholic Church.

They founded and staffed universities and produced extensive theological and practical writings (Le Goff, 1980; Boyle 1981). The Dominicans emphasized the importance of critical thinking, self-examination, and moral development (Hinnebusch 1966; Lesnick 1989).

Because of the strong association of the Franciscan Order with nature and animals, we would naturally expect more-environmental attitudes in regions with stronger Franciscan exposure. Therefore, we initially hypothesized that higher historical exposure to the Franciscan Order would be associated with more pro-environmental attitudes, while higher historical exposure to the Dominican Order would be associated with less pro-environmental attitudes. However, our final results state the opposite.

3 Literature Review

A similar study was conducted by Arruñada and López-Manuel (2024), who examined the effects of the two main Mendicant orders of the Late Middle Ages - the Dominican and the Franciscan. Through their emphasis on guilt, shame, compassion, and analytical thinking, these orders shape outcomes at the cognitive, interpersonal, and institutional levels, which together constitute the "foundations" of impersonal exchange. They further argue that eco-theological beliefs can foster distinct values and cultural norms. For example, at the interpersonal level, "Dominicans promoted guilt as a driver of prosocial behavior toward strangers", fostering a culture of individual accountability and independence, "whereas Franciscans encouraged compassion and shame, which strengthened bonds within the group" (Arruñada and López-Manuel, 2024). Consistent with these differences, the authors document contrasting effects of Dominican and Franciscan influence on impersonal exchange. Thus, their findings supports our results for distinct effects of the two orders on environmental attitudes.

Our study also relates to the broader literature on the role of Medieval Churches and persistance of cultural norms. The Church, as an institution, played a pervasive role in every aspect of life in medieval Europe, shaping people's lives, values, and behavior throughout the Middle Ages (Lynch, 2024). Once a religious institution becomes established, the values and norms it historically imposes tend to persist, as such cultural legacies "leave an imprint on values that endures" over time (Inglehart and Baker, 2000). Although religious institutions have lost some of the formal allegiance of their followers in modern societies, Inglehart and Baker's findings suggest that religiously rooted value orientations continue to influence individuals, as these values become embedded in national cultures and are transmitted through educational institutions and mass media, allowing religiously rooted value orientations to persist at the individual level.

Another important strand of literature relates to the determinants of environmental attitudes. In many literatures it was examined that cultural differences might influence environmental attitudes (Chwialkowska et al., 2020; Khan et al., 2024; Yang et al., 2024; Tam, 2025). Some literature also suggests that Religiously grounded moral worldview of nature is associated with how people understand and relate to the environment (Cohen et al., 1976). Some theoretical models argue that pro-environmental attitudes may be rooted in self-interest and utility maximization. In particular, a longer life expectancy may increase individuals' willingness to support environmental efforts to fight climate change (Mariani et al.,2010). Additionally, importantly for our research, many other literature with similar research interests commonly measure the environemntal attitudes using the survey datasets (EVS) (Gelissen, 2007; Behler, 2025).

Our research is the first to examine the effect of exposure to the two main Mendicant orders of the Medieval Ages - the Franciscan and the Dominican - on contemporary environmental attitudes.

4 Data and Methodology

4.1 Exposure Measure

In our reseach, we assume that people that are living close to the monasteries can get influenced or exposed to the theological beliefs of the monasteries. Thus, the exposure measure is defined as a fraction of a region's historical population living geographically close to a monastic house. To estimate historical exposure to the Franciscan and Dominican orders, we borrow from a lot of datasets. First, we employ the geodataset covering the years between 1216 to 1500 from the Mapping Past Societies (MAPS), which includes information on the geographic locations (latitude and longitude) of the Dominican (870 observations) and Franciscan houses (994 observations) and the name of the houses. The Dominican dataset includes the foundation centuries. The only limitation of this dataset is that it does not include the ending years. After we get the monastery location, we need to locate them within regions. To do so, we use the European Commision system for characterising geographical data - Nomenclature of Territorial Units for Statistics (NUTS), as well as country shapefile. NUTS 1 is for large macro regions, like states, NUTS 2 is for more localized regions within for all of the countries. Since Germany has modified their administrative borders, we rely only on the larger one - NUTS 1 regions. We needed to delete monasteries in east Ukraine, Crimea, and Middle East cities (Jerusalem/Acre), since those monateries lie in the locations that are not in the European Union. There are some monasteries that are a little bit outside the EU; we coerce them to fit in regions, such as Romania or Poland or Greece, because they're pretty close border wise.

Once we habe combined the monastery locations with NUTS and shapefile, our next step is to observe historical population data .

HYDE 3.3 (History Database of the Global Environment) Utrecht University population data captures where people lived historically. It is an estimate of population count and density on regular latitude-longitude grid from the very late Mediaval time to the modern period. Grids are very small squares of 85 km^2, and we have an estimate of population density for each grid.

Similarly, we can think of it not in squares, but in the NUTS regions. By doing similar calculations, we can get the exposure measure for each NUTS 2 region (Figure 3).

We draw a circle around the monastery and all the little squares (grids) that are inside that circle are counted as exposed population. In our research, the exposure measure is defined as the sum of exposed population within a given region as a percentage of the total population for both Franciscan and Dominican monasteries:

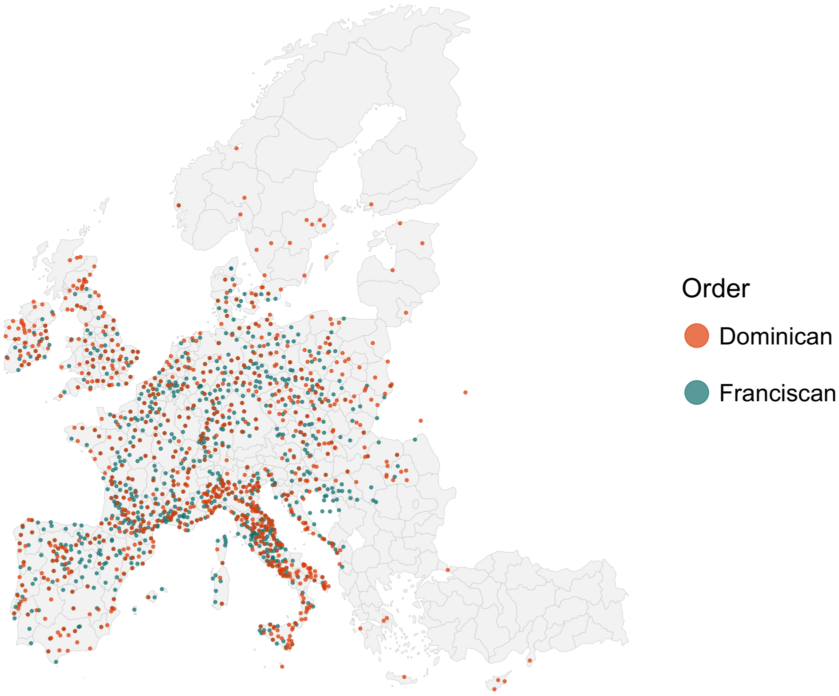


Figure 1: Map of Locations of Mendicant Orders: Franciscan and Dominican Monasteries

$$\text{Exposure} = \frac{\text{Exposed Population}}{\text{Total Population}}$$

Having all the three sources put together, we have Dominican and Franciscan exposure measure for Europe (Figures 4 and 5).

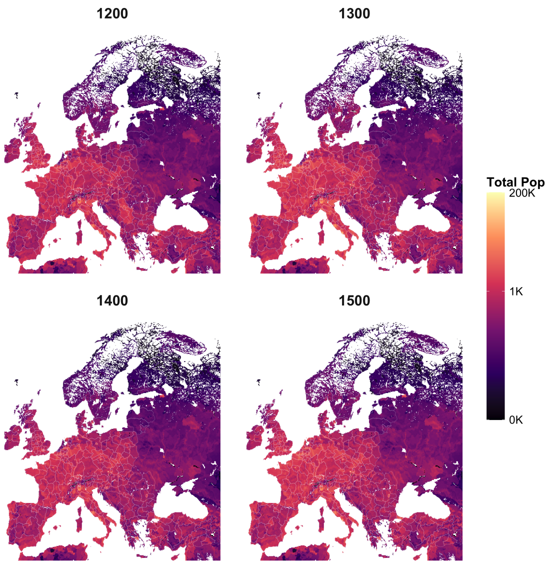
4.2 Environmental Attitudes Measure

To construct our main dependent variable - environmental attitudes, we use the European Values Study (EVS) dataset 2017–2021, which is individual-level survey data on values, beliefs, attitudes, and socioeconomic characteristics across European countries. The dataset covers over 30 European countries (of which 24 are relevant for our research, excluding the countries that are not in NUTS-2). After the data cleaning, the dataset includes 49233 observations, and it includes geographic identifiers for NUTS-1 (Germany) and NUTS-2 regions.

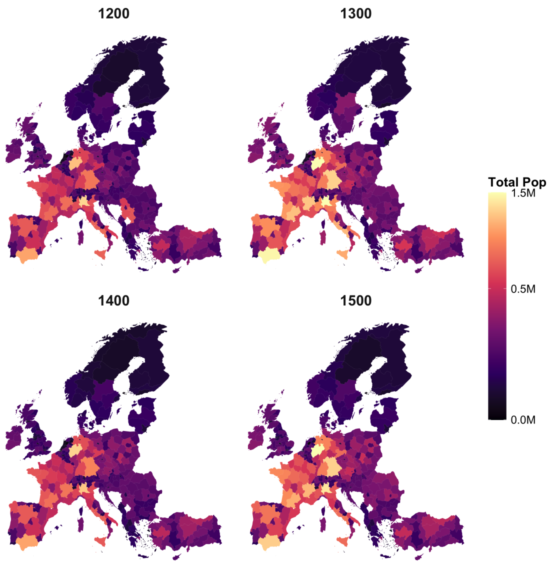
From the EVS dataset, we borrow particularly 6 environmental variables, which are the questions that are related to environmental attitudes. The questions are the following:

- Choosing either "to protect the environment, even if slower economic growth and loss of jobs" or "to have economic growth and newly created jobs, even if environment suffers".

  - Economy and Environment Trade-off question

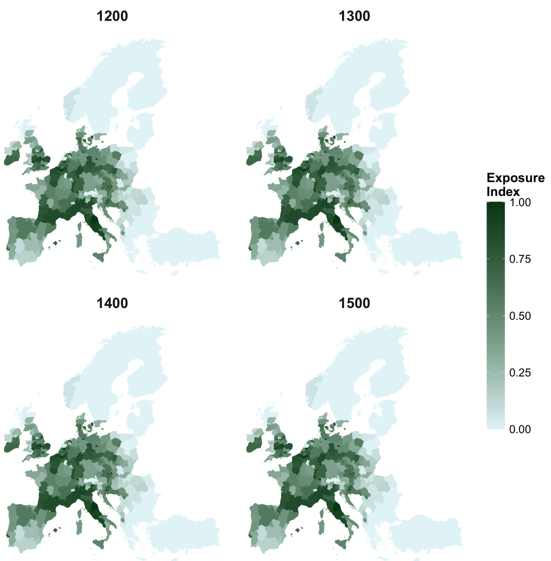


(a) Population densities (grid-level, log scale)

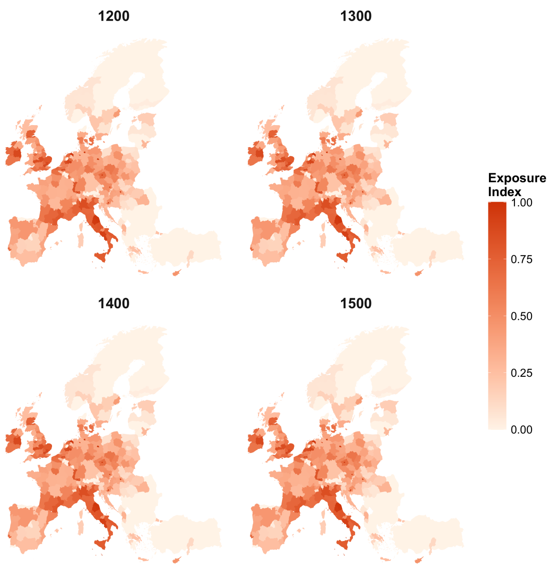


(b) Regional populations (NUTS 2 level)

Figure 2: Population measures from HYDE 3.3



(a) Franciscan exposure



(b) Dominican exposure

Figure 3: Map of exposure to Mendicant Orders in Europe (1200–1500)
normalized index (0 = none, 1 = high exposure)

- "I would give part of my income if I were certain that the money would be used to prevent pollution."
- "It is just too difficult for someone like me to do much about the environment."
- "There are more important things to do in life than protecting the environment."

  - Tells us a bit on people's preferences
- "There is no point in doing what I can for the environment unless others do the same."

  - Network effect and Collective action problem question
- "Many of the claims about environmental threats are exaggerated."

The questions have categorical answers ("strongly agree", "agree", "neither agree nor disagree", "disagree", "strongly disagree", or choosing between the two statements). However, the questions differ in their directional interpretation. Thus, we need to standardize the questions by assigning 0-1 scale per answer for each question, such that the answers indicate the same environmental direction. In the standardized 0-1 scale, 1 indicates the least pro-environmental attitude, and 0 indicates the most pro-environmental attitude. See the Figure 6 for visualization for environmental questions within NUTs 2 regions. The green regions are more pro-environmental, while the yellow regions are less pro-environmental.

Since we have 6 different closely related questions for environmental attitudes, we want to reduce the dimensionality of the data and construct a single Environmental Attitudes Index. Thus, we run a Principal Component Analysis (PCA) to try to extract the main components of the questions (invisible or underlying mechanisms that could drive the answers). We find that the components of the Western European countries behaving more green, while the Eastern European countries behaving more yellowish (see the Figure 7).

5 Results

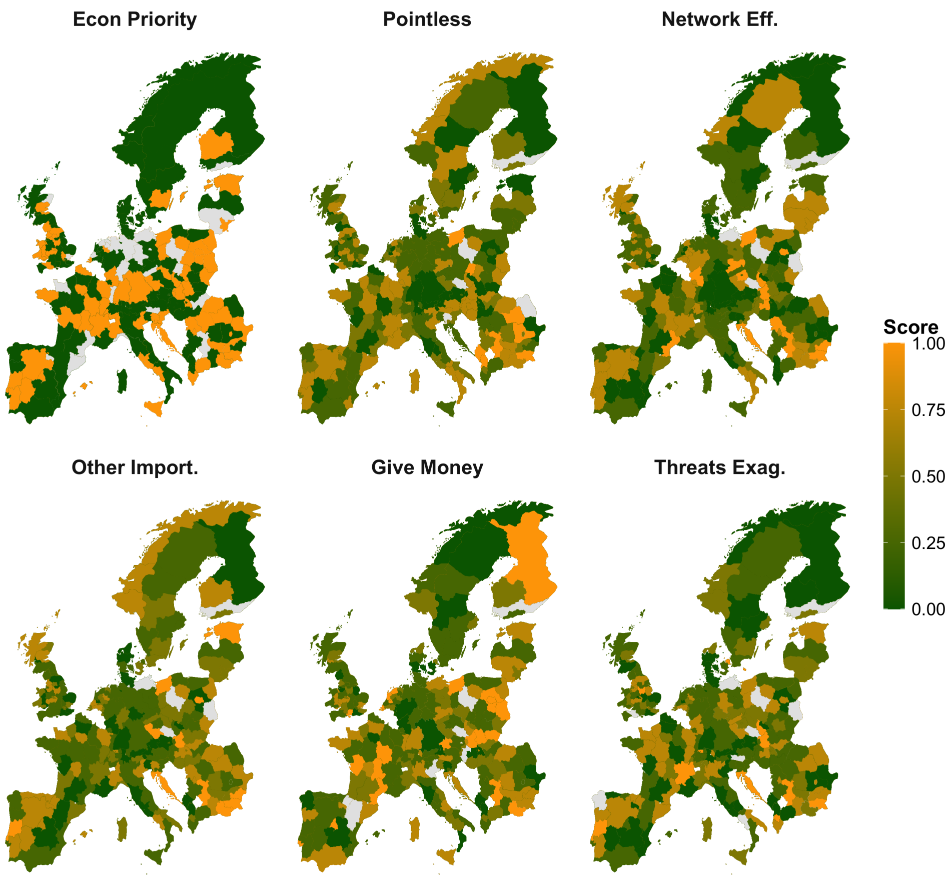


Figure 4: Distribution of the 6 environmental questions across Europe considering the average scores per region.

6 Limitations and Further Research

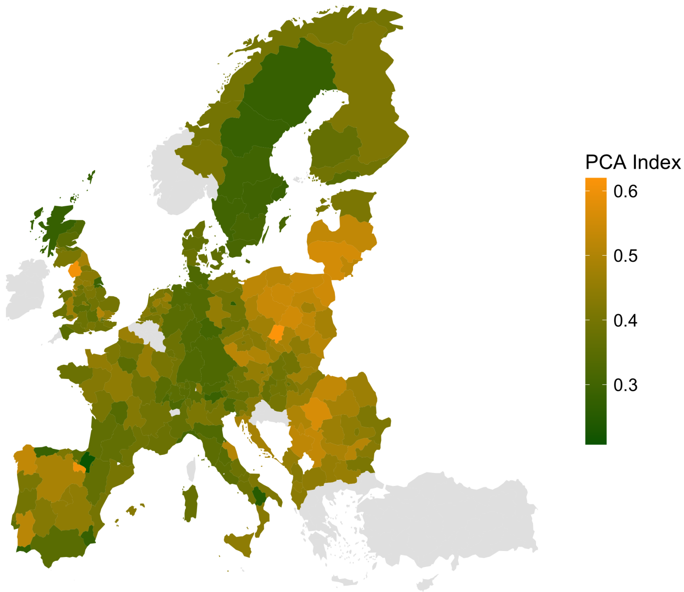


Figure 5: Environmental Index (PCA) across Europe considering the average scores per region.

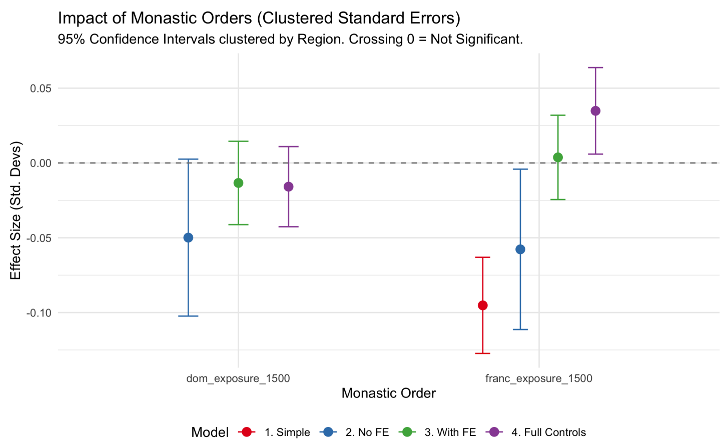


Figure 6: Regressions

7 Conclusion

Regression Results (SE Clustered by Region)

	Model 1: Franciscans Only	Model 2: Fran vs Dom	Model 3: Country FE	Model 4: Full Controls + FE
Franciscan Exposure	-0.095***	-0.058*	0.004	0.035*
	(0.016)	(0.027)	(0.014)	(0.015)
Dominican Exposure		-0.050	-0.013	-0.016
		(0.027)	(0.014)	(0.014)
Gender (Female)				-0.028***
				(0.003)
Age				0.000**
				(0.000)
Education Level				-0.000***
				(0.000)
Income (PPP)				-0.002
				(0.001)
Political (Right)				0.007***
				(0.001)
Town Size				-0.005**
				(0.001)
Socio-Economic Index				-0.001***
				(0.000)
Catholic				0.004
				(0.004)
Protestant				-0.001
				(0.005)
Intercept	0.453***	0.456***	0.439***	0.447***
	(0.009)	(0.009)	(0.004)	(0.011)
Num.Obs.	49233	49233	49233	24379
R2	0.022	0.024	0.099	0.149
R2 Adj.	0.022	0.024	0.099	0.148

* p < 0.05, ** p < 0.01, *** p < 0.001

Figure 7: Regressions table

Individual Environmental Questions (Clustered SEs)

	1. Econ Priority	2. Give Money	3. Pointless	4. Other Import.	5. Network Eff.	6. Threats Exag.
Franciscan Exposure	0.052	0.016	0.034	0.030	0.064**	0.037
	(0.038)	(0.021)	(0.021)	(0.019)	(0.021)	(0.021)
Dominican Exposure	-0.020	-0.026	-0.014	-0.018	-0.039*	-0.017
	(0.037)	(0.021)	(0.018)	(0.016)	(0.016)	(0.015)
Gender (Female)	-0.027***	-0.002	-0.023***	-0.040***	-0.028***	-0.047***
	(0.007)	(0.004)	(0.004)	(0.004)	(0.005)	(0.004)
Age	0.001**	0.001***	0.002***	0.000*	0.001***	0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Education	-0.000**	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Income (PPP)	-0.002	-0.004*	-0.011***	-0.004**	-0.010***	-0.004**
	(0.002)	(0.002)	(0.002)	(0.001)	(0.002)	(0.001)
Socio-Economic Index	-0.002***	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Town Size	-0.003	-0.006**	-0.001	-0.004*	-0.009***	-0.012***
	(0.004)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Political Orientation (Right)	0.011***	0.006***	0.003**	0.004***	0.005***	0.010***
	(0.003)	(0.002)	(0.001)	(0.001)	(0.001)	(0.002)
Catholic	0.007	0.005	0.011*	0.010	0.017*	0.007
	(0.009)	(0.007)	(0.006)	(0.005)	(0.007)	(0.007)
Protestant	-0.003	0.003	-0.007	-0.017**	-0.004	0.008
	(0.013)	(0.012)	(0.006)	(0.005)	(0.006)	(0.008)
Num.Obs.	25427	27212	27344	27370	27343	26823
R2	0.095	0.121	0.140	0.141	0.125	0.121
R2 Adj.	0.094	0.120	0.139	0.139	0.124	0.120

* p < 0.05, ** p < 0.01, *** p < 0.001

Figure 8: Regressions table 2

Robustness Checks: Subsamples

	Main Model (All)	Catholics Only	Exclude Germany
Franciscan Exposure	0.035*	0.023	0.037*
	(0.015)	(0.021)	(0.015)
Dominican Exposure	-0.016	0.004	-0.011
	(0.014)	(0.018)	(0.014)
Num.Obs.	24379	5899	23182
R2	0.149	0.132	0.146
R2 Adj.	0.148	0.126	0.145

* p < 0.05, ** p < 0.01, *** p < 0.001

Figure 9: Regressions table 2

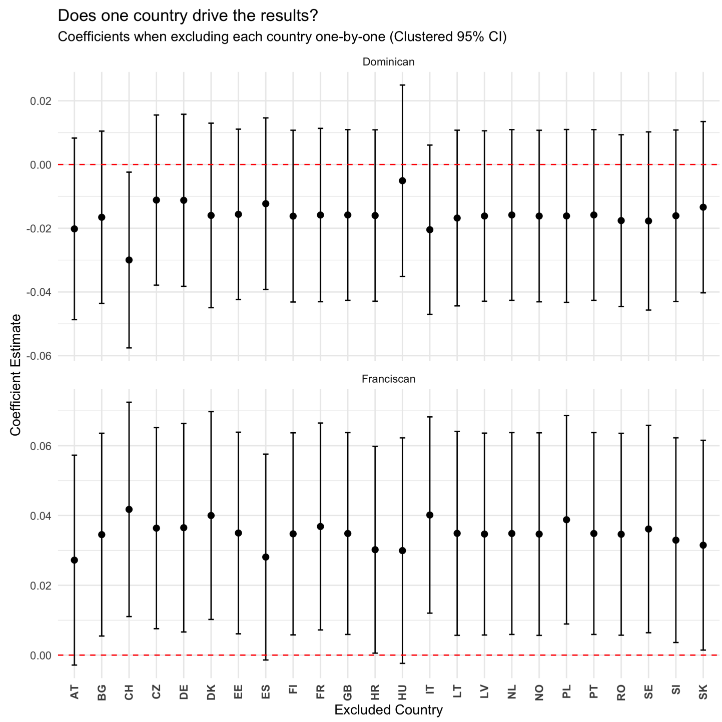


Figure 10: Regressions table 2

Sensitivity Analysis: Coefficients when Dropping Controls

	Full Model	Excl. Gender	Excl. Age	Excl. Education	Excl. Income	Excl. Politics	Excl. Town Size	Excl. Socio-Economic Index	Excl. Religion
Franciscan Exp.	0.035*	0.036*	0.035*	0.035*	0.032*	0.033*	0.012	0.039**	0.034*
Dominican Exp.	-0.016	-0.016	-0.015	-0.017	-0.008	-0.010	-0.010	-0.020	-0.016
Num.Obs.	24379	24379	24434	24438	26795	33560	28220	27286	24379
R2	0.149	0.144	0.149	0.147	0.143	0.135	0.141	0.148	0.149
R2 Adj.	0.148	0.143	0.147	0.145	0.142	0.134	0.139	0.147	0.148

* p < 0.05, ** p < 0.01, *** p < 0.001

Figure 11: Regressions table 2

8 References

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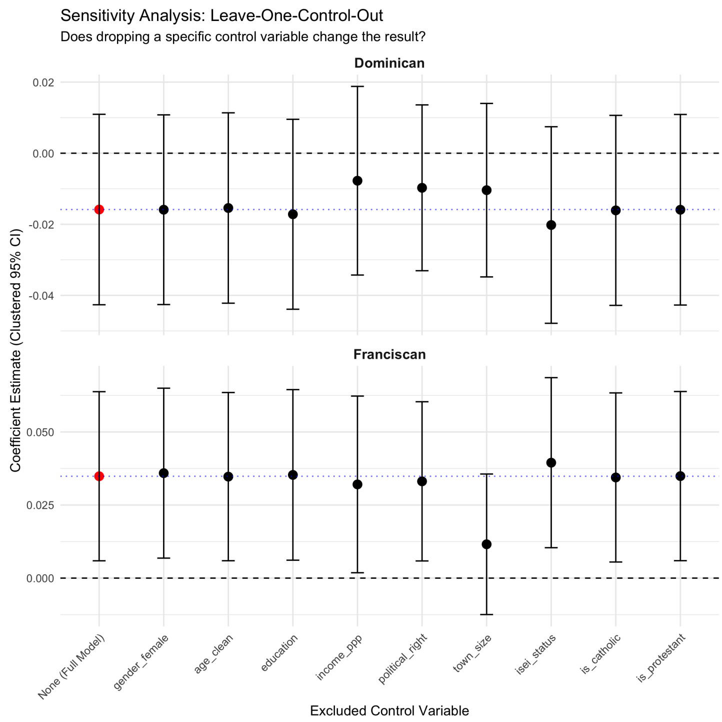

Figure 12: Regressions table 2

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